Air Traffic Passenger Analysis Report

Project Overview

This project analyzes historical air traffic passenger data to uncover trends in airline performance, passenger movement, regional traffic distribution, terminal utilization, and long-term growth patterns. The analysis was conducted using Microsoft SQL Server and covers passenger activity from 1999 to 2026.

Dataset Summary

Metric	Value
Total Records	39,926
Total Airlines	139
Geographic Regions	9
Date Range	1999 – 2026
Total Passengers	1.108 Billion+

The dataset includes passenger traffic categorized by airline, region, terminal, boarding area, activity type, and fare category.

Data Quality Assessment

A comprehensive data quality review was performed before analysis.

Findings

- No missing values in critical columns.
- No negative passenger counts detected.
- Geographic and activity classifications were consistent.
- A small number of duplicate records were identified and documented for review.
- Dataset was deemed suitable for business analysis.

Business Analysis Results

Top Airlines by Passenger Volume

Airline	Passengers
United Airlines	260.0M
United Airlines (Pre 2013)	199.6M
American Airlines	83.5M
SkyWest Airlines	82.7M
Delta Air Lines	71.8M

Insight

United Airlines dominates airport operations and passenger movement, transporting over 260 million passengers during the study period.

Passenger Traffic by Region

Region	Passengers
US	842.8M
Asia	108.9M
Europe	73.0M
Canada	35.0M
Mexico	23.4M

Insight

Domestic US traffic accounts for the majority of passenger activity. Asia is the largest international market, followed by Europe.

Domestic vs International Traffic

Category	Passengers
Domestic	842.8M
International	265.1M

Insight

Approximately 76% of passenger traffic is domestic, while 24% is international.

Passenger Traffic by Terminal

Terminal	Passengers
Terminal 3	454.0M
International	282.0M
Terminal 1	261.1M
Terminal 2	110.8M

Insight

Terminal 3 is the busiest terminal and serves as the primary operational hub.

Passenger Traffic by Boarding Area

Boarding Area	Passengers
F	334.9M
B	149.4M
G	145.5M
A	145.0M

Insight

Boarding Area F handles the largest share of passenger traffic across the airport.

Advanced Analytics

Airline Ranking

United Airlines ranked first among all carriers, followed by its pre-2013 operations, American Airlines, SkyWest Airlines, and Delta Air Lines.

International Airline Leaders

Top international carriers include:

- United Airlines
- Air Canada
- Lufthansa
- British Airways
- EVA Airways
- Cathay Pacific
- Singapore Airlines

Long-Term Growth Analysis

Passenger traffic increased from approximately 21 million passengers in 1999 to over 57 million passengers in 2019.

Growth Trend

- 1999: 20.98M
- 2010: 39.25M
- 2015: 50.06M
- 2019: 57.42M

The airport experienced sustained growth over two decades.

COVID-19 Impact

Year	Passengers	Growth
2019	57.4M	-0.57%
2020	16.4M	-71.41%
2021	24.3M	+48.21%
2022	42.2M	+73.45%
2023	50.1M	+18.79%

Insight

The pandemic caused a historic decline in passenger traffic during 2020. Strong recovery began in 2021 and continued through 2025.

Regional Airline Leaders

Region	Leading Airline
US	United Airlines
Europe	United Airlines
Canada	Air Canada
Middle East	Emirates
Central America	TACA International Airlines
South America	LAN Peru

Terminal Leaders

Terminal	Leading Airline
Terminal 1	Delta Air Lines
Terminal 2	Virgin America
Terminal 3	United Airlines
International Terminal	United Airlines

Key Business Insights

1. United Airlines is the dominant carrier across most regions and terminals.
2. Domestic traffic contributes approximately three-quarters of total passenger volume.
3. Asia represents the largest international passenger market.
4. Terminal 3 is the airport's primary operational hub.
5. Passenger demand showed strong long-term growth before the COVID-19 disruption.
6. Recovery after the pandemic has been substantial, with passenger numbers approaching pre-pandemic levels.
7. International traffic remains a major growth opportunity for airport operators and airlines.

Conclusion

This analysis demonstrates how SQL can be used to transform raw aviation data into actionable business insights. Through data cleaning, aggregation, window functions, trend analysis, and ranking techniques, key drivers of airport performance were identified. The findings highlight the importance of dominant carriers, regional demand patterns, terminal utilization, and long-term passenger growth in supporting strategic airport planning and operational decision-making.